Faculty of Technology and Engineering

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Student ID: Date: / /

Practical Performa

Academic Year	:	2025-26	Semester	:	3rd
Course code	:	CSUC201	Course name	:	Fundamentals of Data Structure and Algorithms

Practical- 10 Hashing

10.1 Hashing

Questions and Answers (Handwritten):

1. Match the **hashing techniques/terms** in **List-I** with their corresponding **properties/outcomes** in **List-II**.

List – I (Hashing Concept)	List – II (Property/Outcome)
(a) Linear Probing	(i) Causes primary clustering
(b) Quadratic Probing	(ii) Eliminates primary clustering but may have secondary clustering
(c) Double Hashing	(iii) Minimizes clustering by using a second hash function
(d) Chaining (Open Hashing)	(iv) Requires linked list for collision resolution
(e) Load Factor (α)	(v) Ratio of number of stored keys to table size

Answer Codes:

(a) → ___

(b) → ___

(c) → ___

(d) → ___

(e) → ___

2. Fill in the blanks (___________) to correctly implement insertion into a hash table using linear probing.

```
#include <iostream>

using namespace std;

const int SIZE = 7;

int hashFunc(int key) {

   return key % SIZE;

}

void insertKey(int table[], int key) {

   int index = hashFunc(key);

   int start = index;

   while (table[index] != -1) {

      // Collision occurred → move to next slot

      index = (index + ________) % SIZE;   // (1)

      if (index == start) {

         cout << "Hash Table is full!" << endl;

         return;

      }

   }

   table[index] = ________;   // } (2)

int main() {

   int table[SIZE];

   for (int i = 0; i < SIZE; i++)

      table[i] = -1;

   insertKey(table, 10);

   insertKey(table, 20);

   insertKey(table, 15);

   for (int i = 0; i < SIZE; i++)

      cout << i << " → " << table[i] << endl;

   return 0;

}
```

3. **Quadratic Probing & Load Factor**

A company uses a hash table of size 11 with hash function: $h(x) = (3x+5) \bmod 11$

Collision resolution: Quadratic probing with $c1=1, c2=3$.

Keys to insert: 20, 34, 45, 70, 56, 90, 12.

(a) Insert the keys step by step. Show the probe sequence for each collision.

(b) What is the **load factor** after all insertions?

(c) Identify any **primary/secondary clustering** patterns.

(d) If the policy is to rehash when load factor > 0.75, does rehashing occur here? If yes, compute the new table size (next prime after doubling).

Grade/Marks

(_____ / 10)

Sign of Lab Teacher with Date